

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Audio Processor
Manufacturer: Audac
Firmware Version: 1.2.0
Model(s): XMP44, ISP40

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.06.0001-b003	1.4.1

Version History

Module Version	Date	Notes
1_0_2_0	11/14/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- MaxFavoriteStations variable must be set accordingly for ISP40 only. Default value is '1'.
MaxFavoriteStations ranges from '1' – '10'.
Example: `InterfaceName.MaxFavoriteStations = '1'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='XMP44')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='XMP44')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 5001, Model='XMP44')</code>

Favorite Station Command

- FavoriteStationUpdate
 - This command queries the device for the latest list of favorite stations and store them into the FavoriteStationResult command.
- FavoriteStationResult
 - This command retruns the result from the FavoriteStationUpdate query based on the selected Slot and Button qualifier. The MaxFavoriteStations variable determines the max number of FavoriteStationResult that can be displayed on the TLP. If the MaxFavoriteStations variable is set to 5, then we can call `InterfaceName.ReadStatus('FavoriteStationResult', {'Slot': '1', 'Button': x})` where x can be 1 to 5 to display 5 results at a time on the TLP.
- FavoriteStationNavigation
 - This command is used to scroll through the results after sending FavoriteStationUpdate.
- FavoriteStationSelect
 - The number selected with this command corresponds to the button in the FavoriteStationResult. It will activate the station with the selected name.

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AutomaticTuning ^{1, 2, 6}	Value 'Up'	Value 'Down'
Qualifier Key 'Slot'	Qualifier Value '1' – '4'	
# AutomaticTuning example InterfaceName.Set('AutomaticTuning', 'Up', {'Slot': '1'})		
Command Band ^{1, 6}	Value 'Toggle'	
Qualifier Key 'Slot'	Qualifier Value '1' – '4'	
# Band example InterfaceName.Set('Band', 'FM', {'Slot': '1'})		
Command FavoriteStationNavigation ⁶	Value 'Up'	Value 'Down'
	Value 'Page Down'	
Qualifier Key 'Slot'	Qualifier Value '1' – '4'	
# FavoriteStationNavigation example InterfaceName.Set('FavoriteStationNavigation', 'Up', {'Slot': '1'})		
Command FavoriteStationSelect ⁶	Value 1 – 10	
Qualifier Key 'Slot'	Qualifier Value '1' – '4'	
# FavoriteStationSelect example InterfaceName.Set('FavoriteStationSelect', 10, {'Slot': '1'})		
Command FavoriteStationUpdate ⁶	Value None	
Qualifier Key 'Slot'	Qualifier Value '1' – '4'	
# FavoriteStationUpdate example InterfaceName.Set('FavoriteStationUpdate', None, {'Slot': '1'})		
Command OutputGain ⁴	Value -92 to 8 in steps of 1 dB	
Qualifier Key	Qualifier Value	

'Slot'	'1' – '4'		
# OutputGain example InterfaceName.Set('OutputGain', 8, {'Slot': '1'})			
Command SelectPreset ^{1, 2, 6}	Value '1' – '10'		
Qualifier Key 'Slot'	Qualifier Value '1' – '4'		
# SelectPreset example InterfaceName.Set('SelectPreset', '1', {'Slot': '1'})			
Command Transport ^{3, 6}	Value 'Play' 'Begin' 'Fast Forward'	Value 'Stop' 'Next' 'Fast Rewind'	Value 'Pause' 'Previous'
Qualifier Key 'Slot'	Qualifier Value '1' – '4'		
# Transport example InterfaceName.Set('Transport', 'Play', {'Slot': '1'})			
Command TuningFrequency ^{1, 2, 5, 6}	Value 87 to 108 in steps of 0.05 MHz		
Qualifier Key 'Slot'	Qualifier Value '1' – '4'		
# TuningFrequency example InterfaceName.Set('TuningFrequency', 108, {'Slot': '1'})			

¹ This will work for slots outfitted with DSP40 and DMP40 only² This will work for slots outfitted with TSP40 and TMP40 only.³ This will work for slots outfitted with MSP40 and MMP40 only.⁴ Every module has its own range. Valid ranges include:

Module FMP40: -92 to 8 dB

Module DMP40: -35 to 8 dB

Module TMP40: -55 to 8 dB

Module IMP40: -92 to 8 dB

⁵ This varies in step size⁶ Available for ISP40 only

Status Available

For all commands except FavoriteStationResult, call Update to receive the latest status. ConnectionStatus and FavoriteStationResult do not support the Update function. ConnectionStatus is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command Band ^{1, 7}	Value 'FM'	Value 'DAB'
Qualifier Key 'Slot'	Qualifier Value '1' – '4'	
# Band examples InterfaceName.Update('Band', {'Slot': '1'}) Value = InterfaceName.ReadStatus('Band', {'Slot': '1'}) InterfaceName.SubscribeStatus('Band', None, FeedbackHandler)		
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)		
Command DeviceType	Value 'String'	
# DeviceType examples InterfaceName.Update('DeviceType') Value = InterfaceName.ReadStatus('DeviceType') InterfaceName.SubscribeStatus('DeviceType', None, FeedbackHandler)		
Command FavoriteStationResult ⁶	Value 'String'	
Qualifier Key 'Slot'	Qualifier Value '1' – '4'	
Qualifier Key 'Button'	Qualifier Value 1 to 10 in steps of 1	
# FavoriteStationResult examples Value = InterfaceName.ReadStatus('FavoriteStationResult', {'Slot': '1', 'Button': 10}) InterfaceName.SubscribeStatus('FavoriteStationResult', None, FeedbackHandler)		

Command	Value		
OutputStatus ^{1,7}	'Stereo'	'Mono'	
Qualifier Key	Qualifier Value		
'Slot'	'1' – '4'		
# OutputStatus examples InterfaceName.Update('OutputStatus', {'Slot': '1'}) Value = InterfaceName.ReadStatus('OutputStatus', {'Slot': '1'}) InterfaceName.SubscribeStatus('OutputStatus', None, FeedbackHandler)			
Command	Value		
OutputGain ⁴	-92 to 8 in steps of 1 dB		
Qualifier Key	Qualifier Value		
'Slot'	'1' – '4'		
# OutputGain examples InterfaceName.Update('OutputGain', {'Slot': '1'}) Value = InterfaceName.ReadStatus('OutputGain', {'Slot': '1'}) InterfaceName.SubscribeStatus('OutputGain', None, FeedbackHandler)			
Command	Value	Value	Value
PlayStatus ^{3,7}	'Playing'	'Paused'	'Stopped'
Qualifier Key	Qualifier Value		
'Slot'	'1' – '4'		
# PlayStatus examples InterfaceName.Update('PlayStatus', {'Slot': '1'}) Value = InterfaceName.ReadStatus('PlayStatus', {'Slot': '1'}) InterfaceName.SubscribeStatus('PlayStatus', None, FeedbackHandler)			
Command	Value		
SignalStatus ^{1,2,7}	0 to 100 in steps of 1%		
Qualifier Key	Qualifier Value		
'Slot'	'1' – '4'		
# SignalStatus examples InterfaceName.Update('SignalStatus', {'Slot': '1'}) Value = InterfaceName.ReadStatus('SignalStatus', {'Slot': '1'}) InterfaceName.SubscribeStatus('SignalStatus', None, FeedbackHandler)			
Command	Value		
SongName ⁶	'String'		
Qualifier Key	Qualifier Value		
'Slot'	'1' – '4'		
# SongName examples InterfaceName.Update('SongName', {'Slot': '1'}) Value = InterfaceName.ReadStatus('SongName', {'Slot': '1'}) InterfaceName.SubscribeStatus('SongName', None, FeedbackHandler)			
Command	Value		
StationName ⁶	'String'		
Qualifier Key	Qualifier Value		
'Slot'	'1' – '4'		

<pre># StationName examples InterfaceName.Update('StationName', {'Slot': '1'}) Value = InterfaceName.ReadStatus('StationName', {'Slot': '1'}) InterfaceName.SubscribeStatus('StationName', None, FeedbackHandler)</pre>	
Command TuningFrequency ^{1, 2, 5, 7}	Value 87 to 108 in steps of 0.05 MHz
Qualifier Key 'Slot'	Qualifier Value '1' – '4'
<pre># TuningFrequency examples InterfaceName.Update('TuningFrequency', {'Slot': '1'}) Value = InterfaceName.ReadStatus('TuningFrequency', {'Slot': '1'}) InterfaceName.SubscribeStatus('TuningFrequency', None, FeedbackHandler)</pre>	

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² This will work for slots outfitted with the TSP40 and TMP40 only.

³ This will work for slots outfitted with the MSP40 and MMP40 only.

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Module DMP40: -35 to 8 dB

Module TMP40: -55 to 8 dB

Module IMP40: -92 to 8 dB

⁵ This varies in step size

⁶ Available for ISP40 only.

⁷ Not available for ISP40.

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 19200

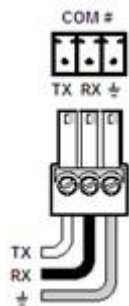
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	3	RxD
RxD	←	2	TxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	5001
Logon Credentials Supported:	No
Multi-Connection Capabilities:	No
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Automatic Tuning Down Slot 1	# D001 web SFSDN1 0 U \x0D\x0A
Automatic Tuning Down Slot 4	# D001 web SFSDN4 0 U \x0D\x0A
Automatic Tuning Up Slot 1	# D001 web SFSUP1 0 U \x0D\x0A
Automatic Tuning Up Slot 4	# D001 web SFSUP4 0 U \x0D\x0A
Band Toggle Slot 1	# D001 web SSBND1 0 U \x0D\x0A
Band Toggle Slot 4	# D001 web SSBND4 0 U \x0D\x0A
Output Gain 8 Slot 1	# D001 web SOG1 0 U \x0D\x0A
Output Gain 8 Slot 4	# D001 web SOG4 0 U \x0D\x0A
Output Gain -92 Slot 1	# D001 web SOG1 100 U \x0D\x0A
Output Gain -92 Slot 4	# D001 web SOG4 100 U \x0D\x0A
Select Preset 1 Slot 1	# D001 web SELPR1 1 U \x0D\x0A
Select Preset 1 Slot 4	# D001 web SELPR4 1 U \x0D\x0A
Select Preset 10 Slot 1	# D001 web SELPR1 10 U \x0D\x0A
Select Preset 10 Slot 4	# D001 web SELPR4 10 U \x0D\x0A
Transport Begin Slot 1	# D001 web SPGTST1 0 U \x0D\x0A
Transport Begin Slot 4	# D001 web SPGTST4 0 U \x0D\x0A
Transport Fast Forward Slot 1	# D001 web SPFFW1 0 U \x0D\x0A
Transport Fast Forward Slot 4	# D001 web SPFFW4 0 U \x0D\x0A
Transport Fast Rewind Slot 1	# D001 web SPFRW1 0 U \x0D\x0A
Transport Fast Rewind Slot 4	# D001 web SPFRW4 0 U \x0D\x0A
Transport Next Slot 1	# D001 web SPNEXT1 0 U \x0D\x0A
Transport Next Slot 4	# D001 web SPNEXT4 0 U \x0D\x0A
Transport Pause Slot 1	# D001 web SPPAUS1 0 U \x0D\x0A
Transport Pause Slot 4	# D001 web SPPAUS4 0 U \x0D\x0A
Transport Play Slot 1	# D001 web SPPLAY1 0 U \x0D\x0A
Transport Play Slot 4	# D001 web SPPLAY4 0 U \x0D\x0A
Transport Previous Slot 1	# D001 web SPPREV1 0 U \x0D\x0A
Transport Previous Slot 4	# D001 web SPPREV4 0 U \x0D\x0A
Transport Stop Slot 1	# D001 web SPSTOP1 0 U \x0D\x0A
Transport Stop Slot 4	# D001 web SPSTOP4 0 U \x0D\x0A
Tuning Frequency 108.0 Slot 1	# D001 web SFREQ1 108.0 U \x0D\x0A
Tuning Frequency 108.0 Slot 4	# D001 web SFREQ4 108.0 U \x0D\x0A
Tuning Frequency 87.0 Slot 1	# D001 web SFREQ1 87.0 U \x0D\x0A
Tuning Frequency 87.0 Slot 4	# D001 web SFREQ4 87.0 U \x0D\x0A

Appendix B. Query Commands

Band Slot 1	# D001 web GBND1 0 U \x0D\x0A
Band Slot 4	# D001 web GBND4 0 U \x0D\x0A
Device Type	# D001 web GTPS 0 U \x0D\x0A
Output Gain Slot 1	# D001 web GOG1 0 U \x0D\x0A
Output Gain Slot 4	# D001 web GOG4 0 U \x0D\x0A
Output Status Slot 1	# D001 web GSTST1 0 U \x0D\x0A
Output Status Slot 4	# D001 web GSTST4 0 U \x0D\x0A
Play Status Slot 1	# D001 web GPSTAT1 0 U \x0D\x0A
Play Status Slot 4	# D001 web GPSTAT4 0 U \x0D\x0A
Signal Status Slot 1	# D001 web GSIGS1 0 U \x0D\x0A
Signal Status Slot 4	# D001 web GSIGS4 0 U \x0D\x0A
Song Name Slot 1	# D001 web GSON1 0 U \x0D\x0A
Song Name Slot 4	# D001 web GSON4 0 U \x0D\x0A
Station Name Slot 1	# D001 web GSTN1 0 U \x0D\x0A
Station Name Slot 4	# D001 web GSTN4 0 U \x0D\x0A
Tuning Frequency Slot 1	# D001 web GFREQ1 0 U \x0D\x0A
Tuning Frequency Slot 4	# D001 web GFREQ4 0 U \x0D\x0A